# Abhiram Paidimarri

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## PROFESSIONAL SUMMARY

AI Engineer with 5+ years of experience building production-grade GenAI systems across enterprise environments. Currently at Microsoft engineering multi-agent orchestration pipelines using Azure AI Foundry and the Microsoft Agent Framework, including ARIA — an autonomous ICM routing system using nested agents for validation and refinement. Specialized in LLMs, RAG pipelines, agentic workflows, vector databases, and full-stack development across AWS and Azure.

## TECHNICAL SKILLS

**Languages:** Python, JavaScript (ES6+), TypeScript, Java, SQL, JSON
**Frontend:** React.js, Redux, TypeScript, Bootstrap, Material-UI, Tailwind, Fluent UI
**Backend & APIs:** Node.js, Express.js, FastAPI, Flask, RESTful APIs, OAuth 2.0, JWT
**GenAI & Agentic:** Microsoft Agent Framework, M365 Agents, Azure AI Foundry, LangChain, LangGraph, AutoGen, Crew AI, HuggingFace, OpenAI GPT-4/4o, Claude, Gemini Pro, Groq, RAG, Mem0, GitHub Copilot, Vector & Graph DBs: FAISS, Pinecone, Qdrant, Neo4j
**Databases:** PostgreSQL, MongoDB, Redis, SQL Server
**Cloud & DevOps**: AWS (Lambda, EC2, S3, CloudWatch), Azure (App Services, Blob Storage, ADF, Foundry, Alerts), ADO Docker, GitLab, CI/CD, GitHub

## WORK EXPERIENCE

**Microsoft (Contract) | AI Engineer:** **June 2025 – Present**

**Responsibilities: Project ARIA — Multi-Agent Orchestration for Automated ICM Analysis**

- Architected and implemented ARIA, a multi-agent orchestration system built using the Microsoft Agent Framework to automate complex ICM (Incident and Case Management) query analysis.
- Designed a hierarchical workflow of analysis agents**,** review agents**,** and a final validation agent, enabling nested reasoning and cross-verification of outputs.
- Built an orchestrator agent that dispatches incoming ICM queries to specialized analysis agents, aggregates their results, and routes them to review agents for accuracy checks.
- Implemented a final adjudication layer where a validation agent determines correctness; if inconsistencies or errors are detected, the orchestrator triggers corrective re-analysis loops automatically.
- Developed Python-based backend logic and APIs to integrate ARIA into the CAIN ecosystem, enabling reliable multi-step reasoning and ensuring high-precision incident insights.

**Responsibilities: Project CAIN**

- Redesigned and implemented the front-end of the Cain application, creating a new user interface similar to Microsoft 365 Copilot, utilizing TypeScript and Fluent UI.
- Designed and prototyped the user experience using Flutter to create a responsive and intuitive design.
- Developed new APIs and refactor existing ones on the Python FastAPI backend for the Cain application's V2, integrating and orchestrating AI Foundry agents to automate complex workflows and tasks.
- Fine-tuned AI prompts and applied a Retrieval-Augmented Generation (RAG) framework to enhance the accuracy and relevance of search results, ensuring the application was grounded in authoritative data sources.

- Collaborated with cross-functional teams to define project requirements and deliver requirements.

**BayouHub | GenAI Developer:** **Jan 2025 – May 2025**
**Responsibilities**

- Developed an intelligent scraping system to enhance over 30,000 business listings for SEO purposes by automating the extraction of key contact information like official emails, websites, and detailed descriptions. This system also implemented advanced deduplication, city/state-based filtering to efficiently avoid redundant data.
- Built a powerful browser automation agent using custom prompts and multi-LLM support from platforms like OpenAI, Gemini, and Groq to autonomously navigate diverse websites and collect specific business data.
- Leveraged a sophisticated Retrieval-Augmented Generation (RAG) framework with a Pinecone vector database and Neo4j knowledge graph to proactively identify and reuse semantically similar business data, thereby significantly reducing redundant scraping across multiple listings.
- Integrated the LangChain framework to programmatically manage complex prompt templates and structure agent responses, ensuring consistent and uniform data formatting across all collected records.
- Enabled robust, real-time data collection with comprehensive error handling and checkpoints, supporting resumable scraping workflows to prevent data loss.
- Output structured and enriched results in clean CSVs, which were directly used to aid in SEO optimization and content completeness for the entire dataset.

**University of Louisiana at Lafayette | AI Engineer** **Sep 2023 – Dec 2024**
**Responsibilities**

- Developed an AI-powered summarization pipeline using OpenAI Whisper and GPT-3.5 to transcribe lecture recordings into accurate, time-stamped text and concise summaries.
- Implemented LangChain pipelines to generate contextual multiple-choice quizzes and highlight key learning outcomes, improving academic engagement.
- Integrated the solution into the Moodle LMS, enabling real-time access for students and faculty.
- Designed a filter-based UI using React to allow users to navigate content by course, topic, and date.
- Enabled educators to export auto-generated questions and summaries, supporting test creation and planning.
- Achieved a 50% reduction in content review time and a 30% increase in student engagement based on analytics.

**Valuebound Pvt Ltd | Software Engineer** **Sep 2021 – July 2023**
**Responsibilities**

- Led the integration of third-party dialers and CRMs into enterprise platforms using OAuth2.0 authentication.
- Designed and deployed a robust microservices architecture with Kafka and RabbitMQ for real-time ingestion and processing of high-volume call data.
- Built secure, scalable REST APIs using Node.js and Express, protected by JWT-based user authentication.
- Implemented Redis caching to improve API performance and reduce database load by 50%.
- Automated data ingestion workflows by uploading structured JSON records into AWS S3.
- Developed and optimized reusable UI components in ReactJS, improving consistency across the ERP interface.

- Achieved 95% test coverage enforced quality through continuous integration pipelines with GitLab CI/CD.

**ParamLogic Pvt Ltd| Software Trainee** **May 2020 – Aug 2021**
**Responsibilities**

- Built a serverless web scraping pipeline using Python, BeautifulSoup, and AWS Lambda running on a daily CloudWatch schedule — replacing manual data entry entirely.
- Synced extracted data to Google Sheets via API in real-time, improving operational efficiency by 100%.
- Stored dependencies in AWS S3 for modularity and documented full workflow for internal team handoff. .

## Projects

**SmartCoder AI — LLM-Powered Code & Test Case Generator: Live Demo, GitHub**
**Environment: Python, Streamlit, OpenAI SDK, REST APIs, GitHub Actions**

- Developed an interactive web application that allows users to generate optimized code and corresponding unit test cases from natural language queries using multiple LLMs (Large Language Models).
- Integrated multiple OpenAI-compatible providers such as Gemini, ChatGPT, Claude, and Groq, with dynamic model selection and support for custom endpoints and model versions.
- Real-time multi-step interaction including planning, code generation, test writing, and execution suggestions.
- Live logging during inference with step-by-step display of LLM reasoning and tool usage.
- Conditional API key handling for both paid and free models using .env and Streamlit Secrets.
- JSON-based interaction model that aligns with OpenAI's function-calling architecture.
- Enabled support for custom LLMs by allowing users to specify base URLs and models.
- Deployed the project on Streamlit, allowing public access with secret-managed API keys, GitHub integration.

**BraedenBot – Website-Integrated AI Chatbot: Live Demo, GitHub**
**Environment: Qdrant-client, Langchain, LangGraph, Fastapi , Streamlit, UV, Pydantic, Beautifulsoup4, Docker**

- End-to-end RAG pipeline. Crawled Braeden's public website with SitemapLoader + BeautifulSoup, chunked pages using RecursiveCharacterTextSplitter, embedded text via Google Gemini Embeddings (google-generativeai Python SDK), and stored vectors in a Qdrant collection for millisecond-level semantic search.
- Conversational retrieval layer. Orchestrated retrieval and response synthesis with LangChain agents (ConversationalRetrievalChain, custom tool definitions) running on Gemini-1.5 Pro; enforced JSON I/O with pydantic schemas to drive a structured chat UX.
- API & UI. Exposed the bot through a lightweight FastAPI endpoint and a Streamlit chat interface featuring left/right bubble styling, session memory, timeouts, and spinners for non-blocking UX.
- DevOps & packaging. Containerized with Docker, managed dependencies (poetry), and automated test/deploy using GitHub Actions; single-command spin-up for local or cloud (Docker Compose on AWS Lightsail).
- Security & observability. Segregated API keys with python-dotenv & Streamlit Secrets, added rate-limit middleware, and instrumented query/latency metrics with Prometheus + Grafana dashboards.
- Modular roadmap. Architecture designed for plug-in retrieval routes—prototype extensions sketched for a Sales Bot (Bid-Matrix & Proposal generation), HR Bot (policy FAQ), Quality Bot (QMS/NCR lookup), and Tech-Support.

## EDUCATION

- University of Louisiana at Lafayette | Master of Science, Informatics — **GPA: 4.0** | Dec 2024
- JB Institute of Engineering and Technology | Bachelor of Technology, Computer Engineering | Jun 2021

## Certifications

- AWS Certified Solutions Architect – *Validation: 8ecffdbbe84e48788641de7d0815d111*
- AWS Certified Cloud Practitioner – *Validation: WSF102Y2TJ1EQBCB*
- Data Science – Coursera
- Leetcode Top 50 SQL Badge - Badges